Maastricht University

Department of Advanced Computing Sciences

MATSE: Operations Research

Lecture 3. More Simplex

[based on slides by: Steven Kelk and Steven Chaplick]

Barbara Franci

Standard LP ... or is it?

Given: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ [standard form]
where, for each entry b_i in b, $b_i \geq 0$, i.e., $b \geq 0$

Find: $x^\star \in \mathbb{R}^n$ where $x^\star = \arg\max\{c^\mathrm{T} x \mid Ax \leq b, x \geq 0\}$

- Maximization
- All decision variables: non-negative (≥ 0)
- All constraints: $\leq$

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$$\left.\begin{array}{l} \bullet \text{ Maximization} \\ \bullet \text{ All decision variables: non-negative } (\geq 0) \\ \bullet \text{ All constraints: } \leq \end{array}\right\} \text{all of these are \textbf{safe}, \textit{without loss of generality} regarding LP}$$

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- Maximization
- All decision variables: non-negative (≥ 0)
- All constraints: $\leq$

all of these are **safe**, *without loss of generality* regarding LP

With this assumption, an initial BFS is easy to find:

- Set the decision variables as non-basic (set $= 0$), and the slack variables as basic.

The Simplex Method

1. Find a BFS to start (just use the slack variables as the basis).
2. Can a currently nonbasic variable help to improve the objective function? (**just check coefficients in row(0)**)

   No? We are done! We have an optimal solution!
3. Pick a nonbasic variable with the greatest potential to improve the objective function, **the entering variable**
4. Use the **minimum ratio test** to identify the **leaving variable**
5. Update the system for the new BFS using Gaussian elimination
6. GOTO 2.

Algebraic vs. Tabular Form

TABLE 4.3 Initial system of equations for the Wyndor Glass Co. problem

(a) Algebraic Form	(b) Tabular Form								
	Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
(0) $\boldsymbol{Z} - 3x_1 - 5x_2 = 0$	Z	(0)	1	−3	−5	0	0	0	0
(1) $x_1 + \boldsymbol{x_3} = 4$	x_3	(1)	0	1	0	1	0	0	4
(2) $2x_2 + \boldsymbol{x_4} = 12$	x_4	(2)	0	0	2	0	1	0	12
(3) $3x_1 + 2x_2 + \boldsymbol{x_5} = 18$	x_5	(3)	0	3	2	0	0	1	18

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(0) $Z - 3x_1 - 5x_2 = 0$	Z	(0)	1	−3	−5	0	0	0	0
(1) $x_1 + x_3 = 4$	x_3	(1)	0	1	0	1	0	0	4
(2) $2x_2 + x_4 = 12$	x_4	(2)	0	0	2	0	1	0	12
(3) $3x_1 + 2x_2 + x_5 = 18$	x_5	(3)	0	3	2	0	0	1	18

Starting BFS: $(0, 0, 4, 12, 18)$

TABLE 4.4 Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side	Ratio
Z	(0)	1	−3	−5	0	0	0	0	
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- Negative value in row(0), so not yet optimal

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  for each row, find the value of entering var (x_2) such that, basic var. $= 0$

  pick the row (and basic variable) with **minimum** value

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x_4	(2)	0	0	$\boxed{2}$	0	1	0	12	$\to \frac{12}{2} = 6 \leftarrow$ minimum
x_5	(3)	0	3	$\boxed{2}$	0	0	1	18	$\to \frac{18}{2} = 9$

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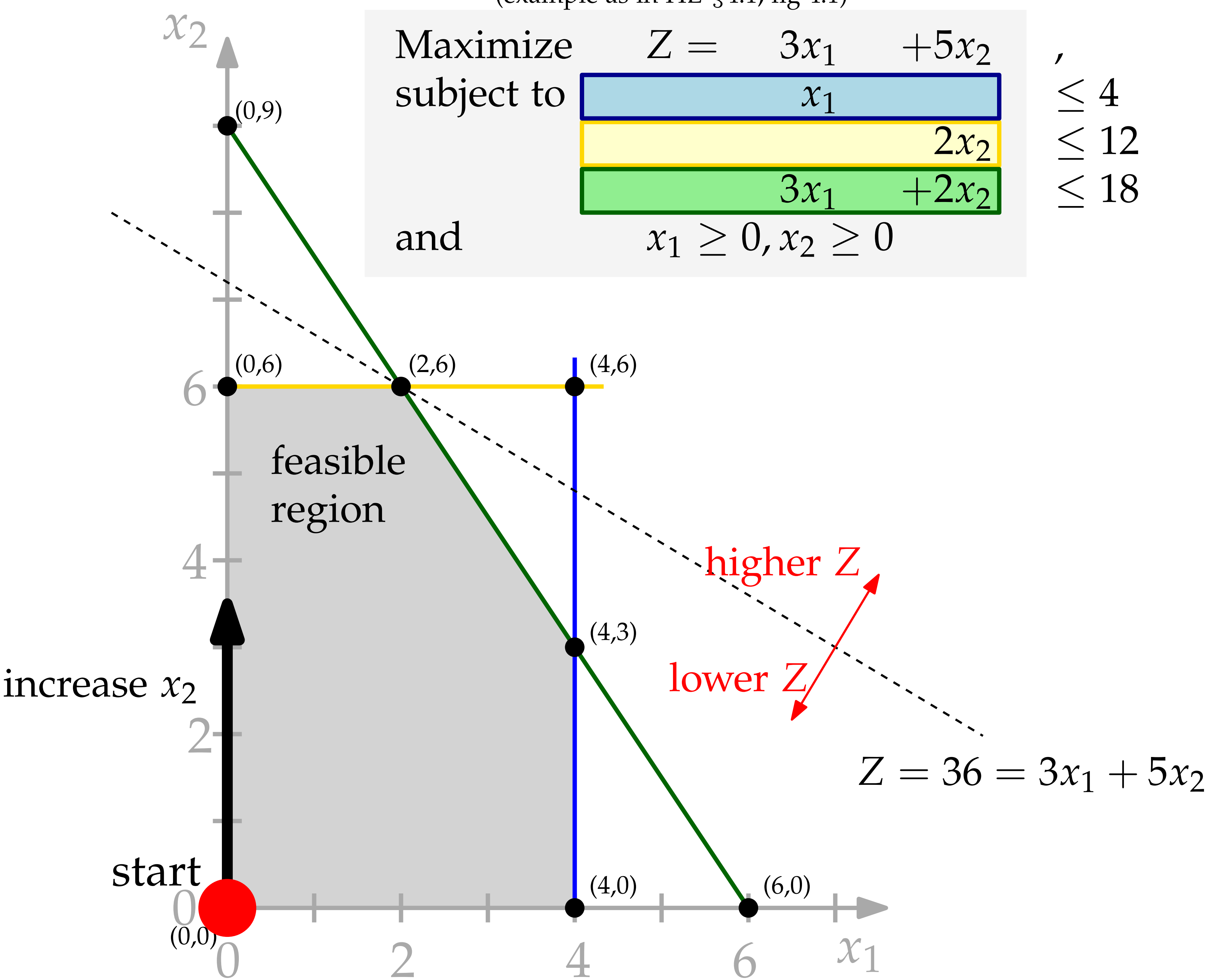
$$\boxed{\text{row(2) gives minimum ratio} \rightsquigarrow x_4 \text{ leaves}}$$

(example as in HL §4.1, fig 4.1)

$$
\begin{array}{llll}
\text{Maximize} & Z = 3x_1 + 5x_2 & , \\
\text{subject to} & x_1 & \leq 4 \\
 & 2x_2 & \leq 12 \\
 & 3x_1 + 2x_2 & \leq 18 \\
\text{and} & x_1 \geq 0, x_2 \geq 0 &
\end{array}
$$



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\text{Maximize} & Z = & 3x_1 & +5x_2 \quad , \\
\text{subject to} & & \boxed{x_1} & \le 4 \\
 & & \boxed{2x_2} & \le 12 \\
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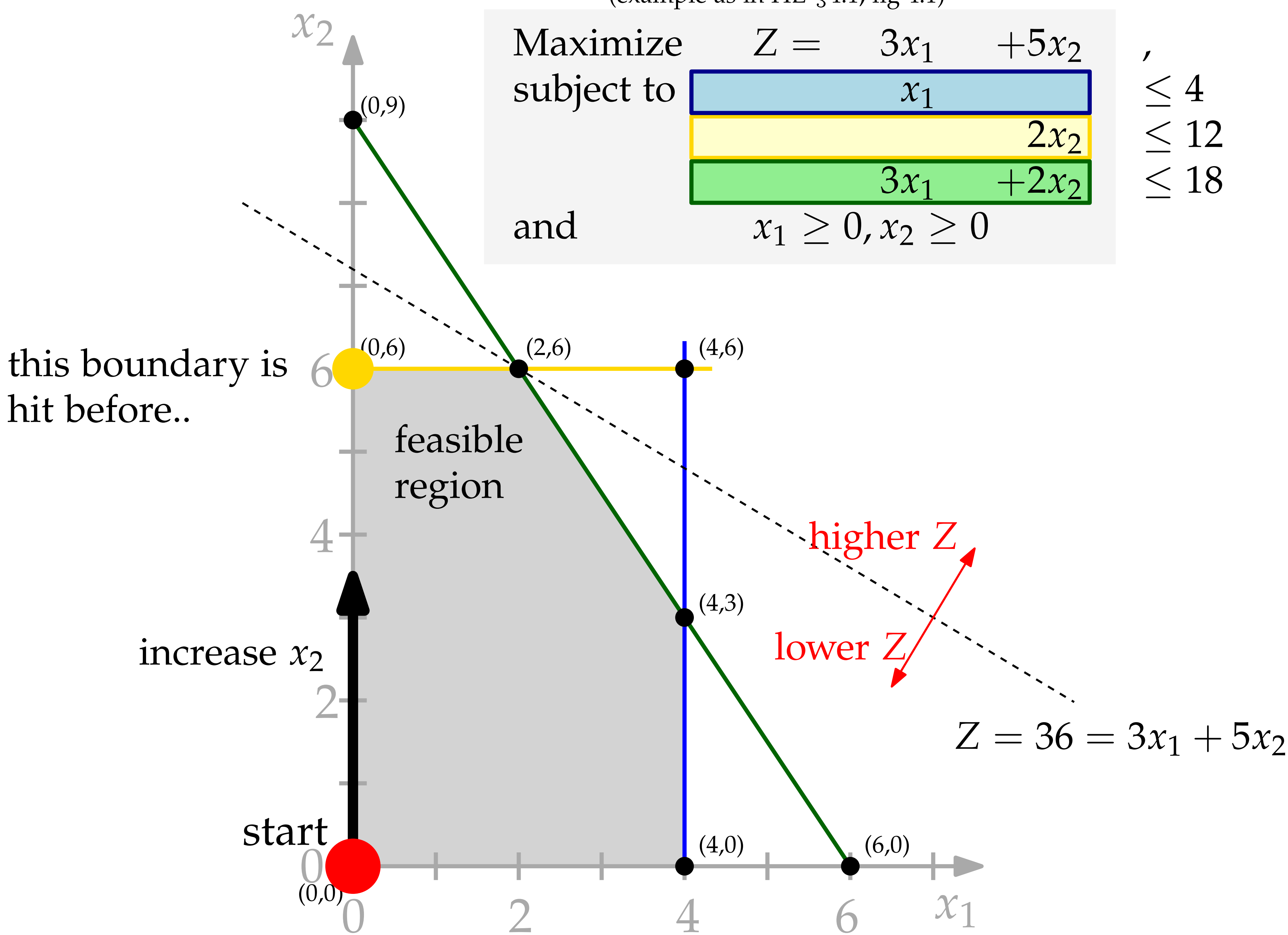
\text{and} & & x_1 \ge 0, x_2 \ge 0 &
\end{array}
$$



(example as in HL §4.1, fig 4.1)

Maximize	$Z = 3x_1 + 5x_2$	,	
subject to	x_1	≤ 4	
	$2x_2$	≤ 12	
	$3x_1 + 2x_2$	≤ 18	
and	$x_1 \geq 0, x_2 \geq 0$		

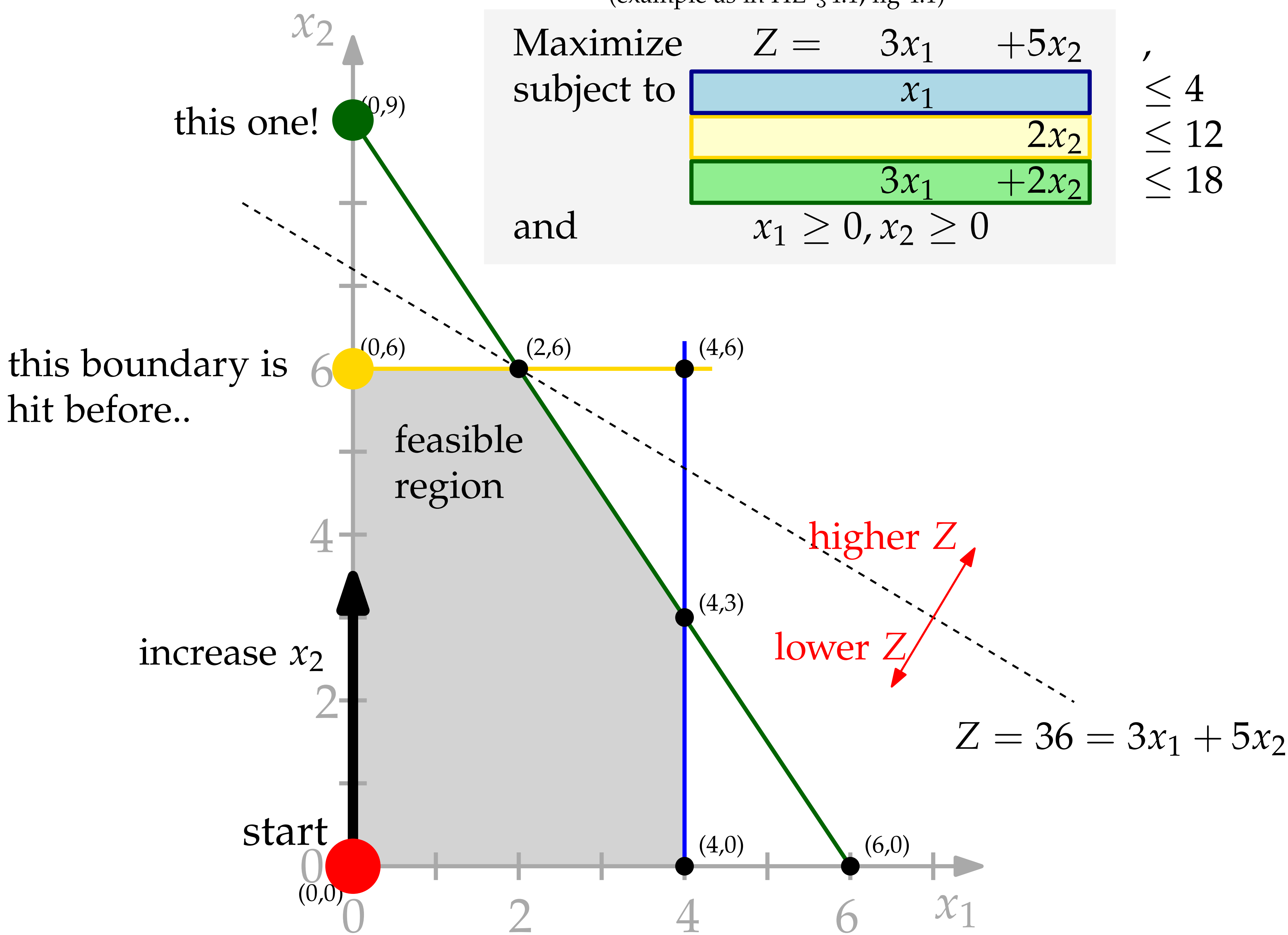


TABLE 4.4 Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side	Ratio
Z	(0)	1	−3	−5	0	0	0	0	
x_3	(1)	0	1	$\boxed{0}$	1	0	0	4	
x_4	(2)	0	0	$\boxed{2}$	0	1	0	12	$12 \rightarrow \frac{12}{2} = 6 \leftarrow$ minimum
x_5	(3)	0	3	$\boxed{2}$	0	0	1	18	$18 \rightarrow \frac{18}{2} = 9$

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):

  $\boxed{-5 \text{ is most negative in row(0)} \rightsquigarrow \text{pick } x_2}$
- Find leaving variable (basic $\rightarrow$ nonbasic): Minimum Ratio Test

  for each row, find the value of entering var (x_2) such that, basic var. $= 0$

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	Z	(0)	1	-3	-5	0	0	0	0	
	x_3	(1)	0	1	0	1	0	0	4	
pivot row	x_4	(2)	0	0	2	0	1	0	12	$\rightarrow \frac{12}{2} = 6 \leftarrow$ minimum
	x_5	(3)	0	3	2	0	0	1	18	$\rightarrow \frac{18}{2} = 9$
					pivot column					

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pivot row	x_4	(2)	0	0	$\boxed{2}$	0	1	0	$12 \rightarrow \frac{12}{2} = 6 \leftarrow$ minimum	
	x_5	(3)	0	3	$\boxed{2}$	0	0	1	$18 \rightarrow \frac{18}{2} = 9$	
					pivot column					

- Next Step: Establish identity 0/1 pattern in the pivot column (1 in pivot row, 0 otherwise).

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Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side	Ratio
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Z	(0)	1		0					
x_3	(1)	0		0					
x_4	(2)	0		1					
x_5	(3)	0		0					

- Next Step: Establish identity 0/1 pattern in the pivot column (1 in pivot row, 0 otherwise).
- Do this via elementary row operations (this will of course change ... many things!)

TABLE 4.5 Simplex tableaux for the Wyndor Glass Co. problem after the first pivot row is divided by the first pivot number

Iteration	Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
0	Z	(0)	1	−3	−5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1						
	x_3	(1)	0						
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	x_5	(3)	0						

TABLE 4.5 Simplex tableaux for the Wyndor Glass Co. problem after the first pivot row is divided by the first pivot number

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	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1						
	x_3	(1)	0						
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0						

TABLE 4.6 First two simplex tableaux for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Z	x_1	x_2	x_3	x_4	x_5	Right Side
0	Z	(0)	1	−3	−5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1	−3	0	0	$\frac{5}{2}$	0	30
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0	3	0	0	−1	1	6

TABLE 4.6 First two simplex tableaux for the Wyndor Glass Co. problem

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	x_3	(1)	0	1	0	1	0	0	4
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	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1	−3	0	0	$\frac{5}{2}$	0	30
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0	3	0	0	−1	1	6

- start the next round: pick an entering variable, then find a leaving variable!

TABLE 4.7 Steps 1 and 2 of iteration 2 for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Z	x_1	x_2	x_3	x_4	x_5	Right Side	Ratio
1	Z	(0)	1	-3	0	0	$\frac{5}{2}$	0	30	
	x_3	(1)	0	1	0	1	0	0	4	$\frac{4}{1} = 4$
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6	
	x_5	(3)	0	3	0	0	-1	1	6	$\frac{6}{3} = 2 \leftarrow$ minimum

- start the next round: pick an entering variable, then find a leaving variable!

TABLE 4.8 Complete set of simplex tableaux for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Z	x_1	x_2	x_3	x_4	x_5	Right Side
0	Z	(0)	1	−3	−5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1	−3	0	0	$\frac{5}{2}$	0	30
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0	3	0	0	−1	1	6
2	Z	(0)	1	0	0	0	$\frac{3}{2}$	1	36
	x_3	(1)	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_1	(3)	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2

Complications (§4.5)

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What if there are unbounded/multiple optima?

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- Tie for Entering Basic Variable:

  multiple nonbasic variables with the same **per unit increase** (most negative coefficient in row(0))

- Tie for Leaving Basic Variable:

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What if there are unbounded/multiple optima?

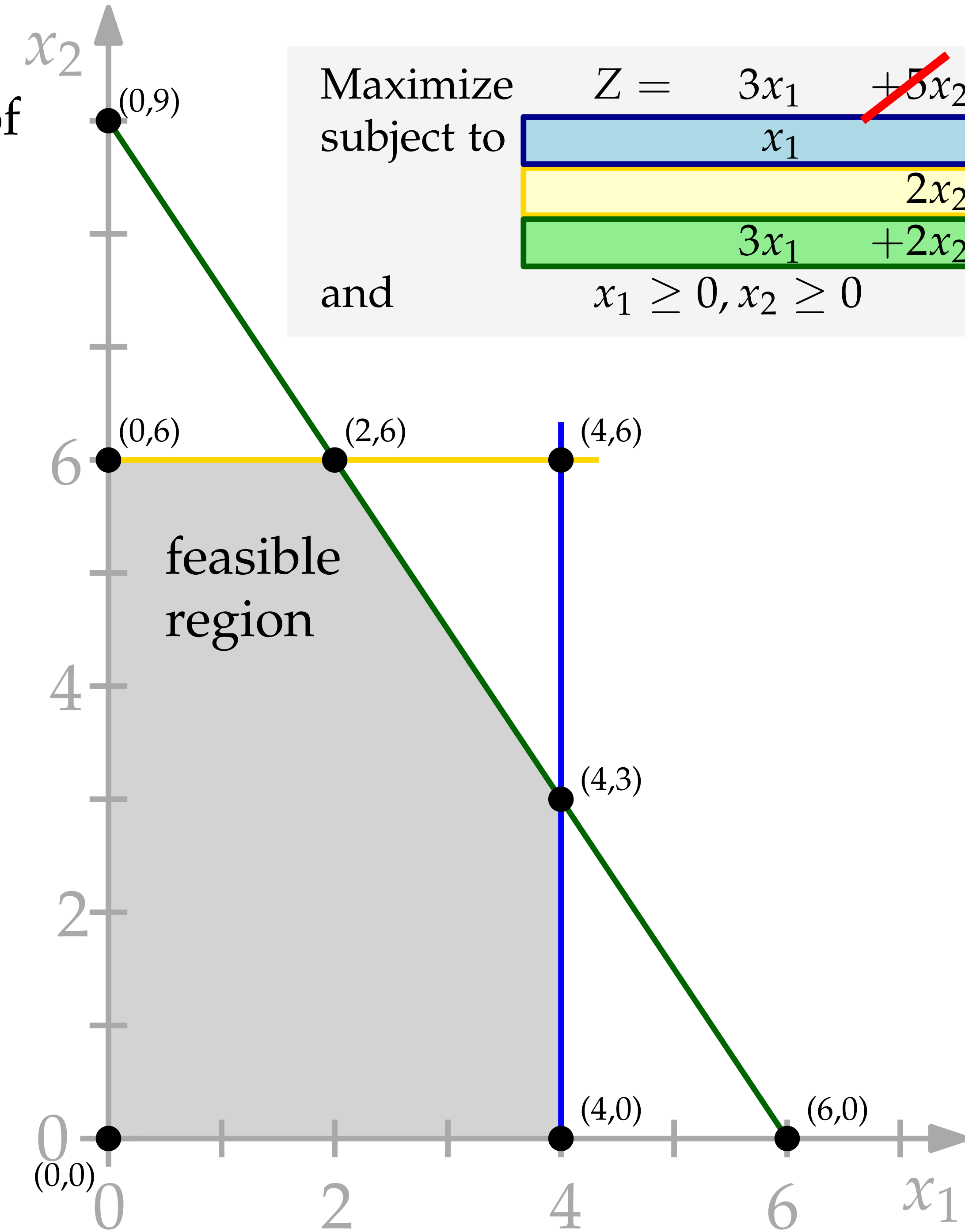
- Tie for Entering Basic Variable:
  multiple nonbasic variables with the same **per unit increase** (most negative coefficient in row(0))

- Tie for Leaving Basic Variable:
  multiple basic variables with the same **Minimum Ratio**

$\leadsto$ Degeneracy: not just theoretical, it happens!

Multiple Optima

- change the coefficient of x_2 to 2 instead of 5.



$$
\begin{array}{lrl}
\text{Maximize} & Z = 3x_1 + \cancel{5}x_2\ 2x_2 & \\
\text{subject to} & \boxed{x_1} & \leq 4 \\
& \boxed{2x_2} & \leq 12 \\
& \boxed{3x_1 + 2x_2} & \leq 18 \\
\text{and} & x_1 \geq 0, x_2 \geq 0 &
\end{array}
$$

Multiple Optima

- change the coefficient of x_2 to 2 instead of 5.
- Now, two optimal CPF solutions. (objective function parallel to green functional constraint)

$$
\begin{array}{llrrl}
\text{Maximize} & Z = & 3x_1 & + \cancel{5x_2}\ 2x_2 & \\
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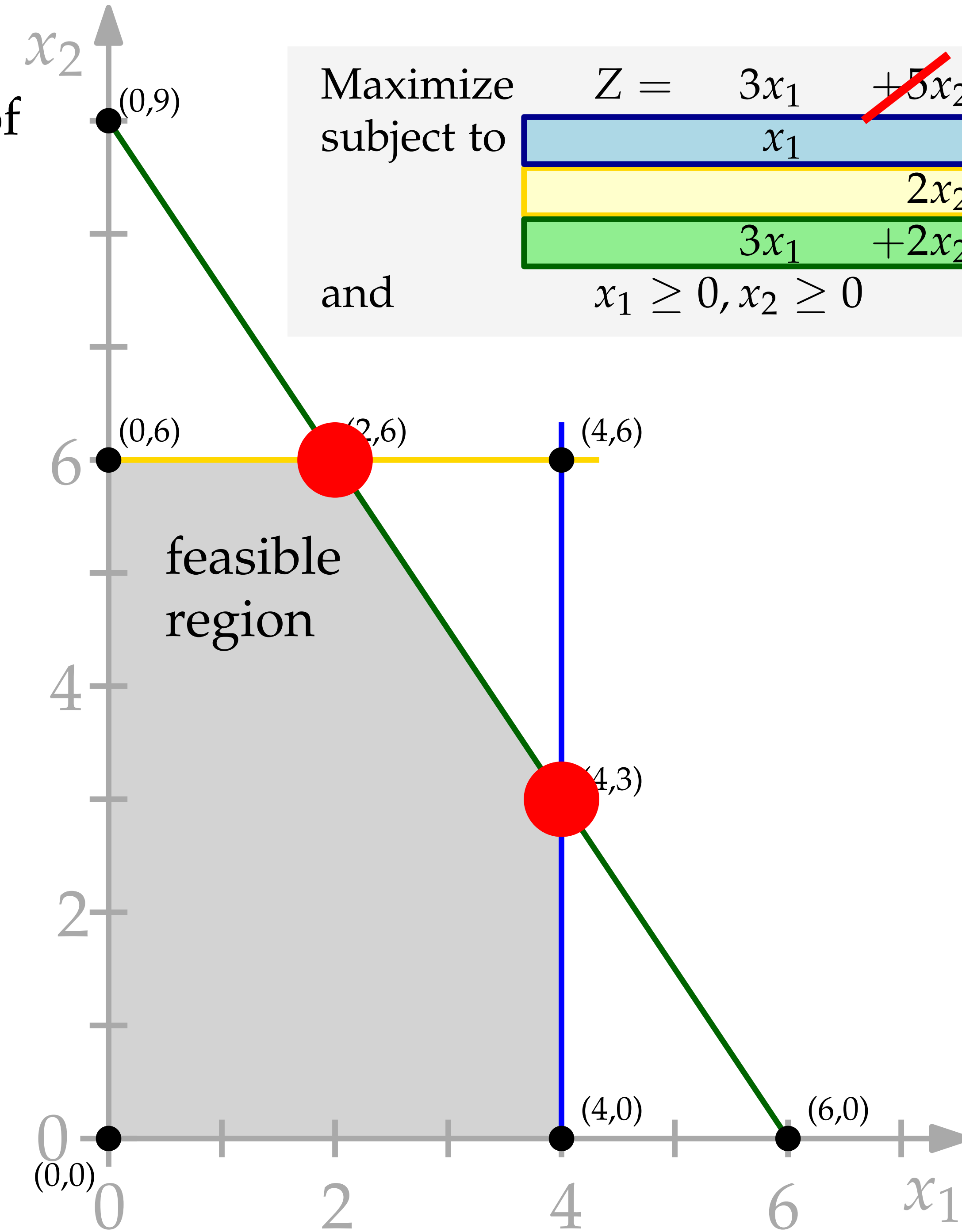
& & & 2x_2 & \leq 12 \\
& & 3x_1 & +2x_2 & \leq 18 \\
\text{and} & & \multicolumn{2}{l}{x_1 \geq 0, x_2 \geq 0} &
\end{array}
$$



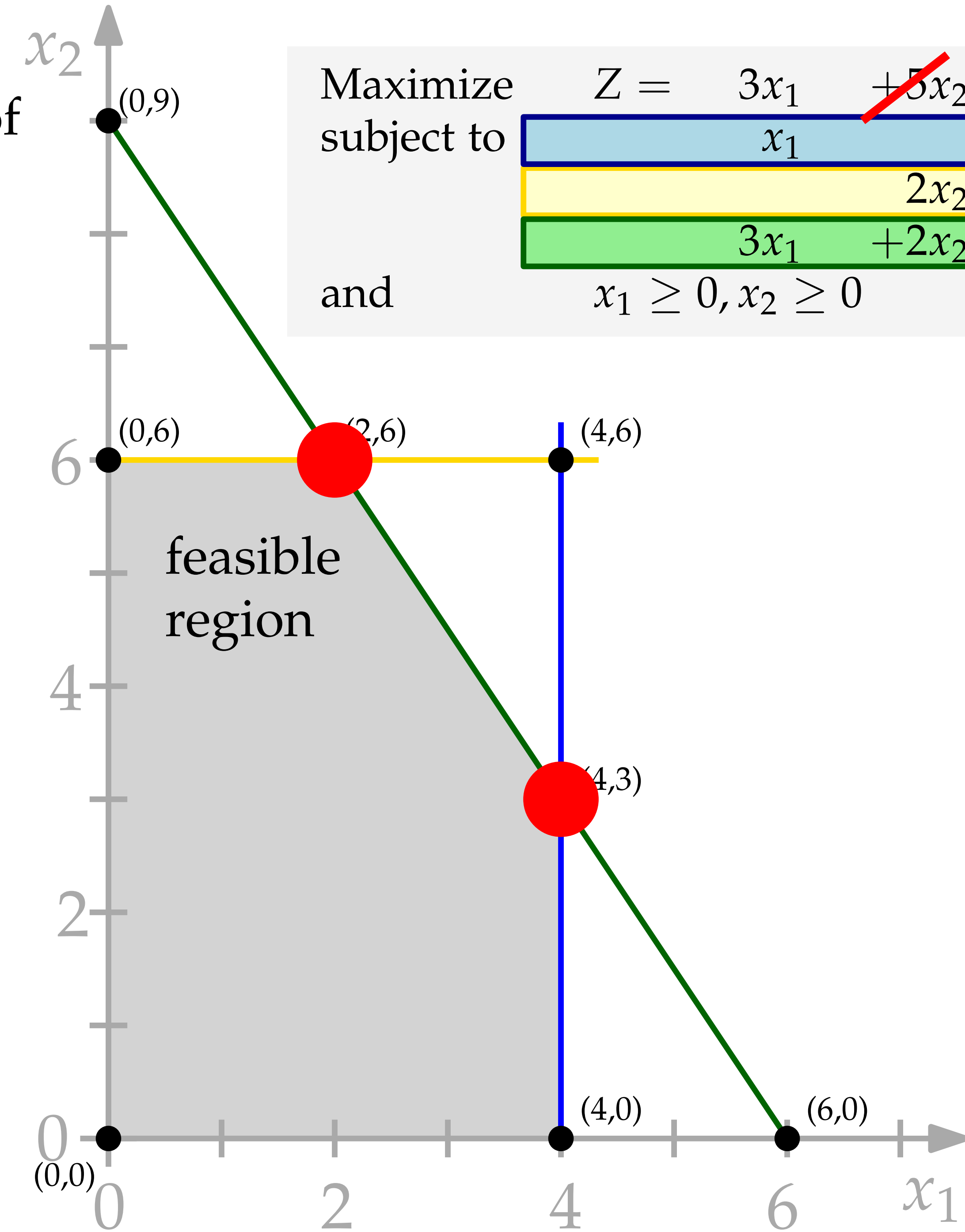
Multiple Optima

- change the coefficient of x_2 to 2 instead of 5.

- Now, two optimal CPF solutions. (objective function parallel to green functional constraint)

- To explore these alternative optima, continue to pivot, choosing a nonbasic variable with a 0 in the top row as the entering basic variable

$$
\begin{array}{llrrl}
\text{Maximize} & Z = & 3x_1 & + \cancel{5x_2}\ 2x_2 & \\
\text{subject to} & & \boxed{x_1} & & \leq 4 \\
 & & & \boxed{2x_2} & \leq 12 \\
 & & \boxed{3x_1} & \boxed{+2x_2} & \leq 18 \\
\text{and} & & x_1 \geq 0, & x_2 \geq 0 &
\end{array}
$$



Multiple Optima

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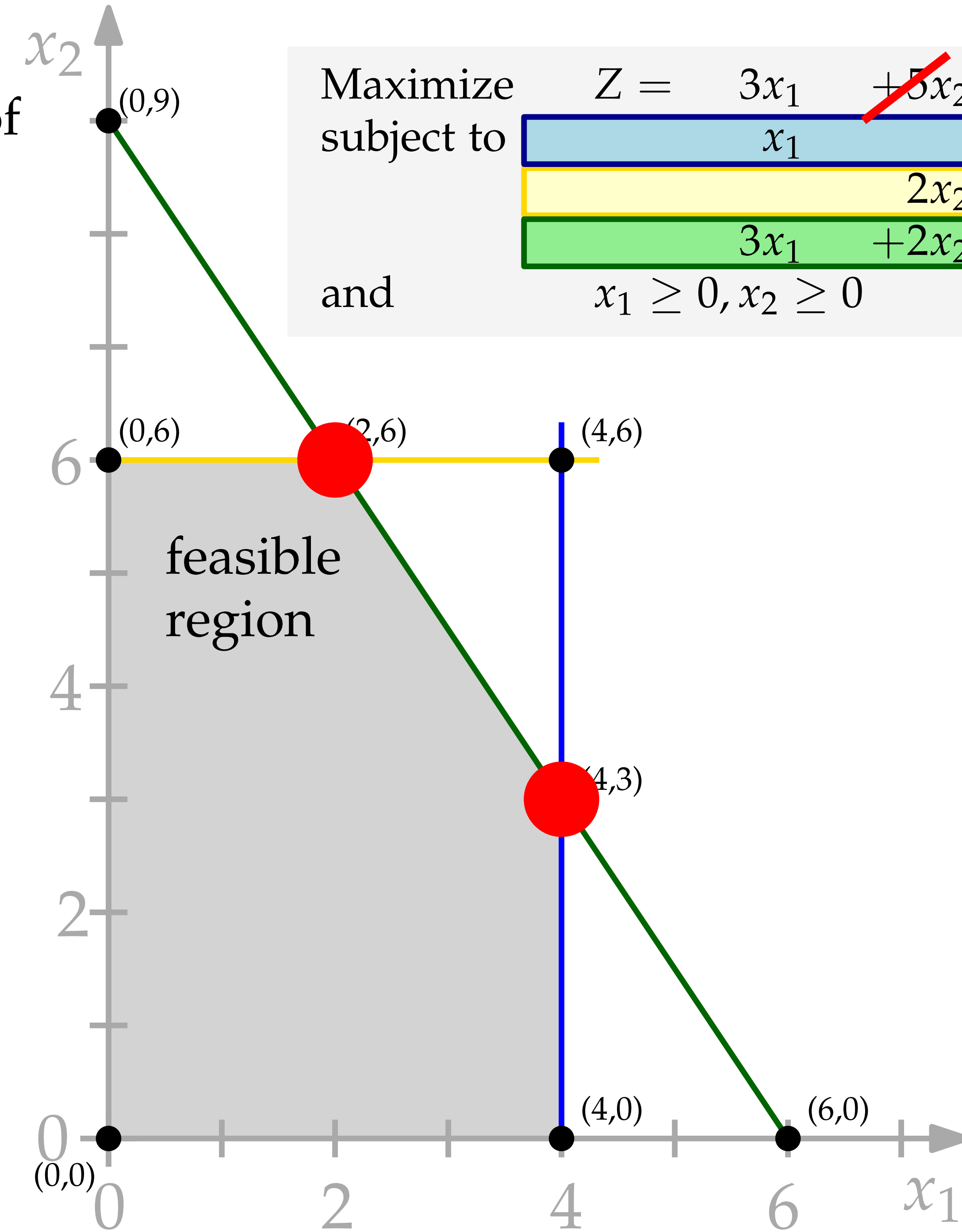
- Now, two optimal CPF solutions. (objective function parallel to green functional constraint)
- To explore these alternative optima, continue to pivot, choosing a nonbasic variable with a 0 in the top row as the entering basic variable

Maximize $Z = 3x_1 + \cancel{5x_2}\ 2x_2$

subject to

$$\begin{aligned} x_1 &\le 4 \\ 2x_2 &\le 12 \\ 3x_1 + 2x_2 &\le 18 \end{aligned}$$

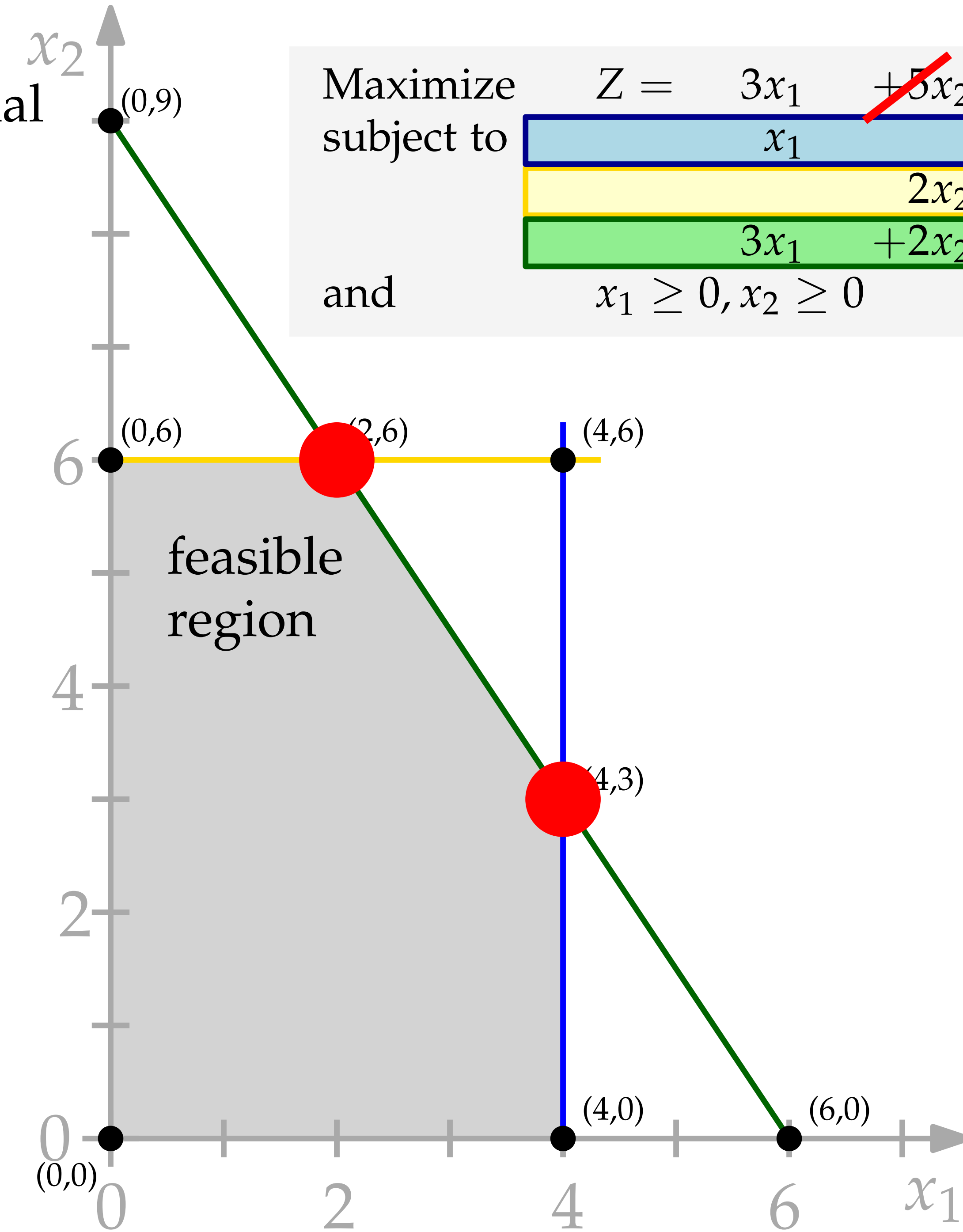
and $x_1 \ge 0, x_2 \ge 0$



Multiple Optima

What are all possible optimal solutions?

$$
\begin{array}{llrrl}
\text{Maximize} & Z = & 3x_1 & + \cancel{5x_2}\ 2x_2 & \\
\text{subject to} & & \boxed{x_1} & & \leq 4 \\
 & & & \boxed{2x_2} & \leq 12 \\
 & & \boxed{3x_1} & \boxed{+2x_2} & \leq 18 \\
\text{and} & & \multicolumn{2}{l}{x_1 \geq 0, x_2 \geq 0} &
\end{array}
$$



Multiple Optima

What are all possible optimal solutions?

$$(x_1, x_2) = w_1(2,6) + w_2(4,3)$$

$$w_1 + w_2 = 1, w_1, w_2 \geq 0$$

$\Rightarrow$ **Convex combinations** of solutions

$$\begin{array}{lll} \text{Maximize} & Z = & 3x_1 + \cancel{5x_2}\ 2x_2 \\ \text{subject to} & & \boxed{x_1} \leq 4 \\ & & \boxed{2x_2} \leq 12 \\ & & \boxed{3x_1 + 2x_2} \leq 18 \\ \text{and} & & x_1 \geq 0, x_2 \geq 0 \end{array}$$

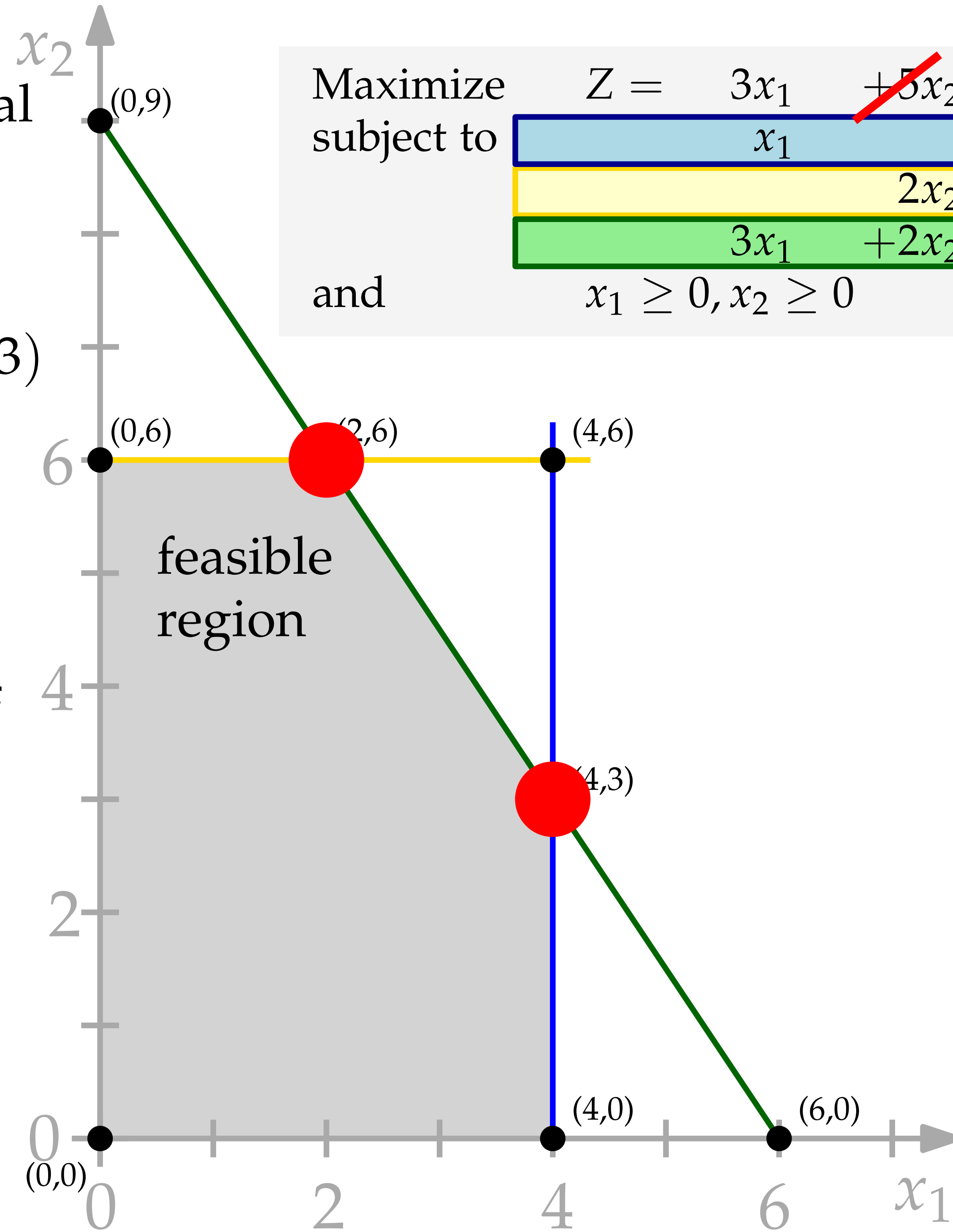


TABLE 4.10 Complete set of simplex tableaux to obtain all optimal BF solutions for the Wyndor Glass Co. problem with $c_2 = 2$

Iteration	Basic Variable	Eq.	Z	x_1	x_2	x_3	x_4	x_5	Right Side	Solution Optimal?
0	Z	(0)	1	−3	−2	0	0	0	0	No
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	3	2	0	0	1	18	
1	Z	(0)	1	0	−2	3	0	0	12	No
	x_1	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	0	2	−3	0	1	6	
2	Z	(0)	1	0	0	**0**	0	1	18	Yes
	x_1	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	0	3	1	−1	6	
	x_2	(3)	0	0	1	$-\frac{3}{2}$	0	$\frac{1}{2}$	3	
Extra	Z	(0)	1	0	0	0	**0**	1	18	Yes
	x_1	(1)	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2	
	x_3	(2)	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2	
	x_2	(3)	0	0	1	0	$\frac{1}{2}$	0	6	

Multiple Optima

- Modern LP solvers can often provide these ☺

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- How to efficiently list/enumerate all the optimal CPF/BFS of an LP is actually a topic of research in its own right!

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- Finding all the optimal CPF/BFS of an LP is mathematically very similar to the problem of finding all the (not necessarily optimal) CPF/BFS of the feasible region.

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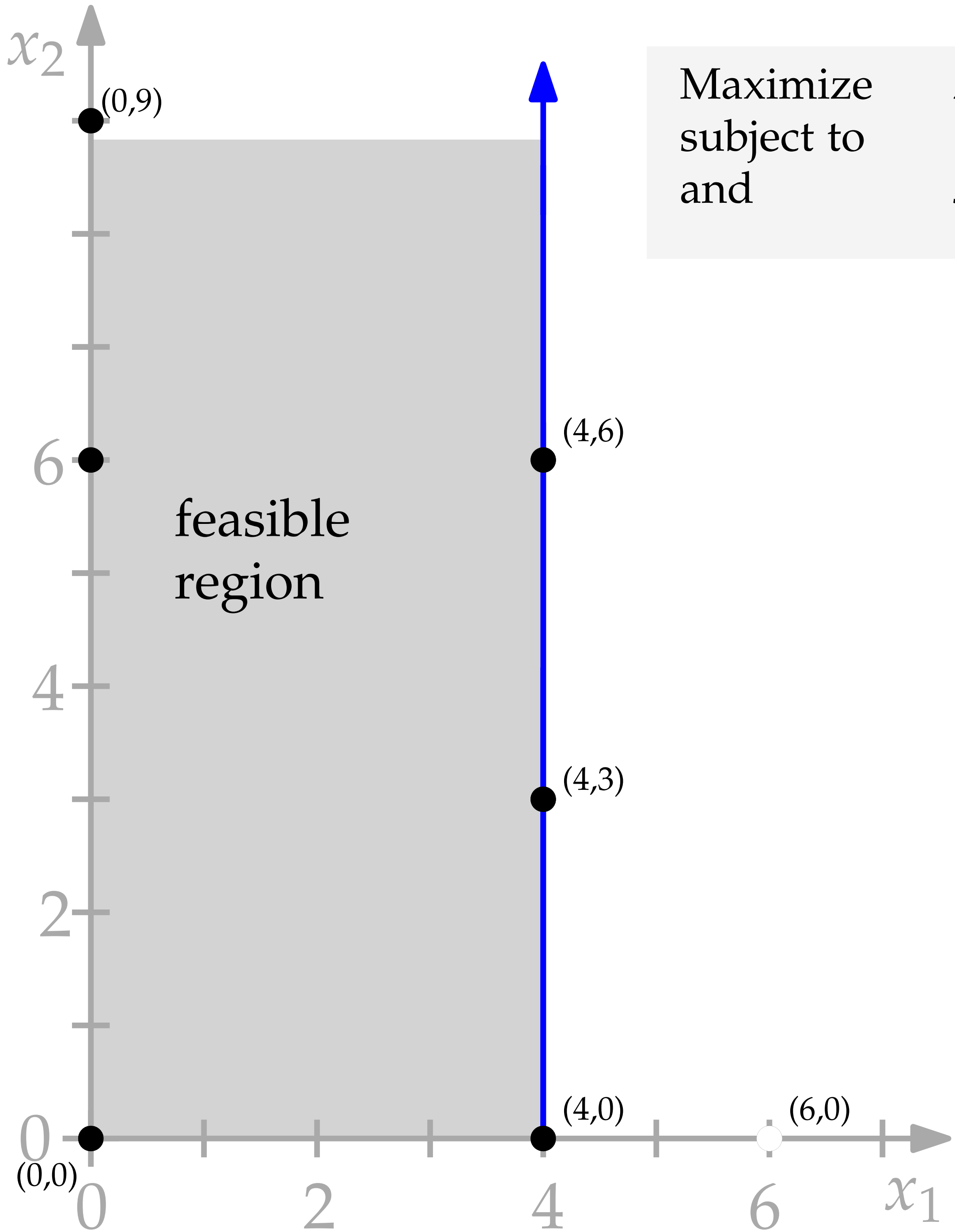
- Finding all the optimal CPF/BFS of an LP is mathematically very similar to the problem of finding all the (not necessarily optimal) CPF/BFS of the feasible region.
- Such exhaustive lists of CPF/BFS can be very useful for characterizing the **entire space** of (optimal) solutions

Unbounded solution



Maximize	$Z =$	$3x_1 + 5x_2$,
subject to		$\boxed{x_1 \leq 4}$
and		$x_1 \geq 0, x_2 \geq 0$

Undefined Minimum Ratio?

TABLE 4.9 Initial simplex tableau for the Wyndor Glass Co. problem without the last two functional constraints

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	Right Side	Ratio
Z	(0)	1	−3	−5	0	0	
x_3	(1)	0	1	$\boxed{0}$	1	4	None

With $x_1 = 0$ and x_2 increasing,
$x_3 = 4 - 1x_1 - 0x_2 = 4 > 0$.

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- The minimum ratio is ∞ (or undefined) ????
- Here, we can increase x_2 arbitrarily (while staying feasible), and as such increase Z arbitarily!
- Thus, unbounded optimal solution
- Yes, we will find this, but first we need to be **adjacent** to such unbounded *face* of the feasible region.

Degeneracy

(1) Tie among entering variables, i.e., multiple nonbasic variables with the same **potential**

(2) Tie among leaving variables: basic variables with the same **minimum ratio**

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don't worry it will be fine, arbitrarily break ties!

(actually, works well in practice...)

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What to do?

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(actually, works well in practice...)

- But, seriously, while it is often claimed to be *rarely* problematic, it is important to be aware of it and have an idea that there are ways to **properly** handle it.

Degeneracy

- A BFS is **degenerate** if one or more of its basic variables is zero, i.e., has RHS $= 0$. Such basic variables are called **degenerate** basic variables.

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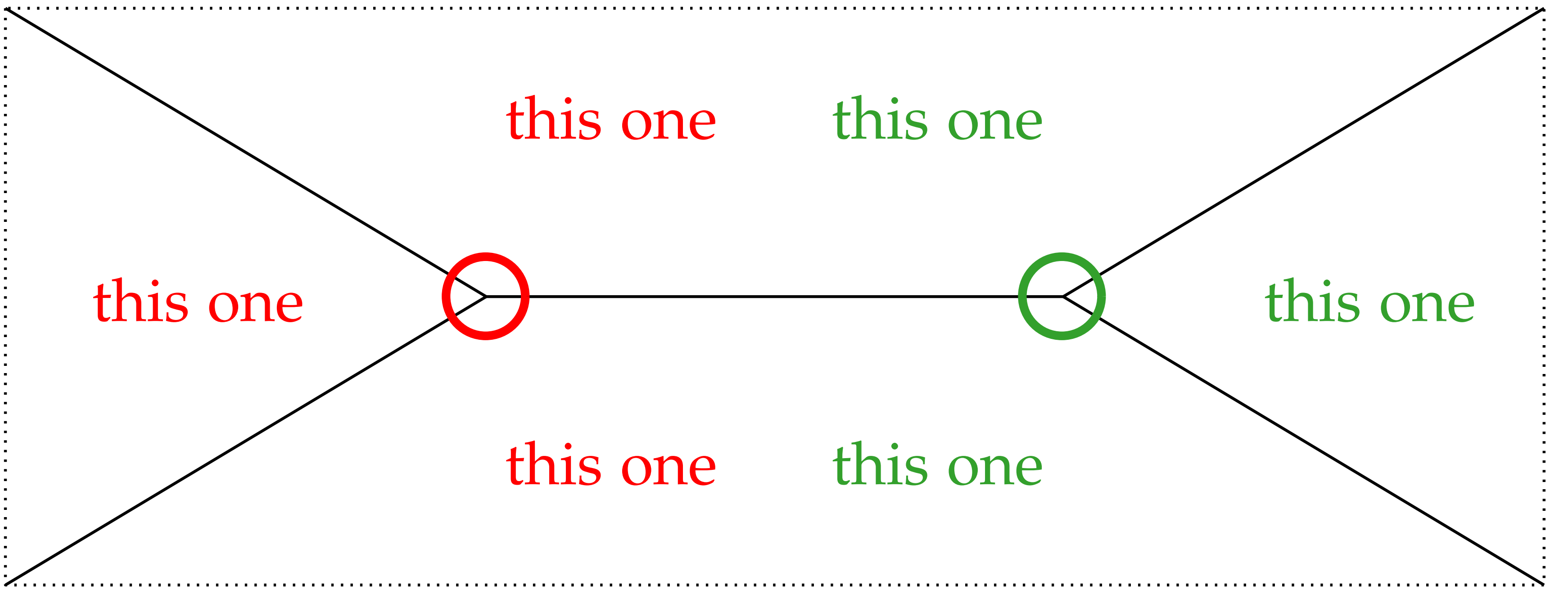
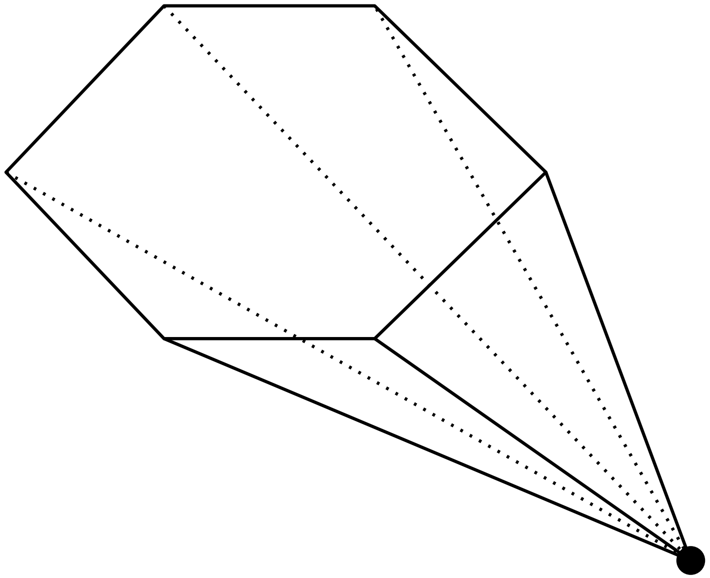
(based on the idea that in n-dimensional space a CPF solution requires n constraints to be tight, i.e. met with equality, at that point. Sure, but...)

Degeneracy



But, what if...

Degeneracy



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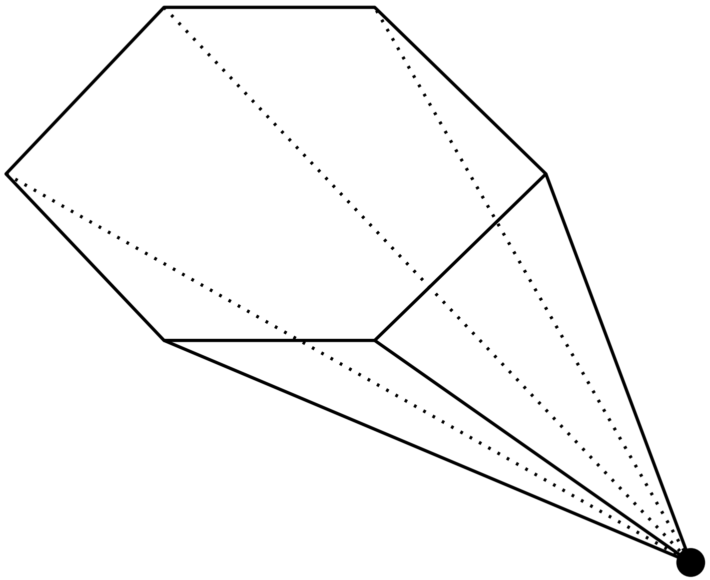
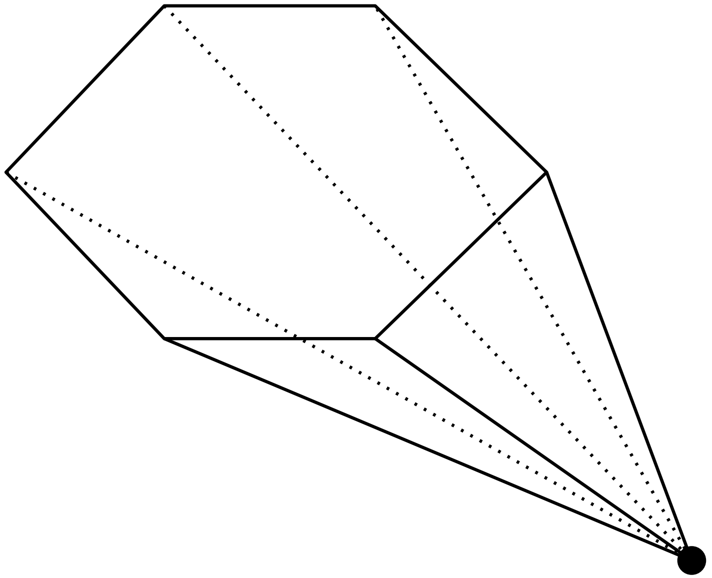


Any **three** of the **six** long *faces* define the corner point!

Degeneracy



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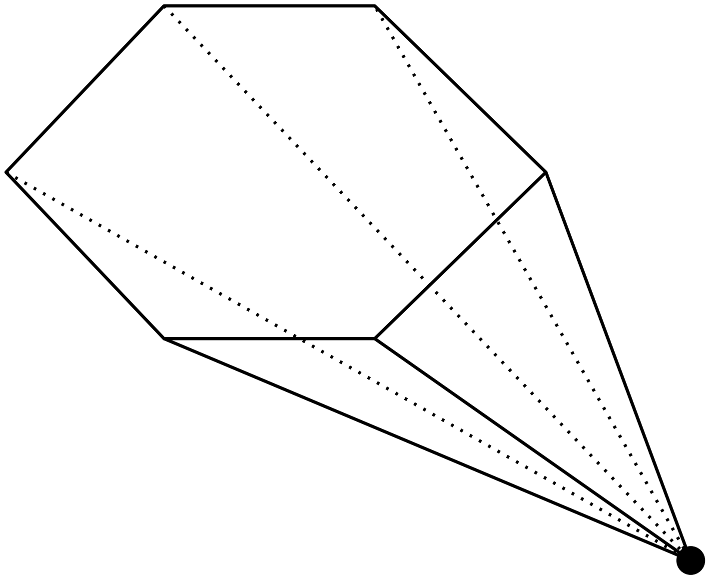

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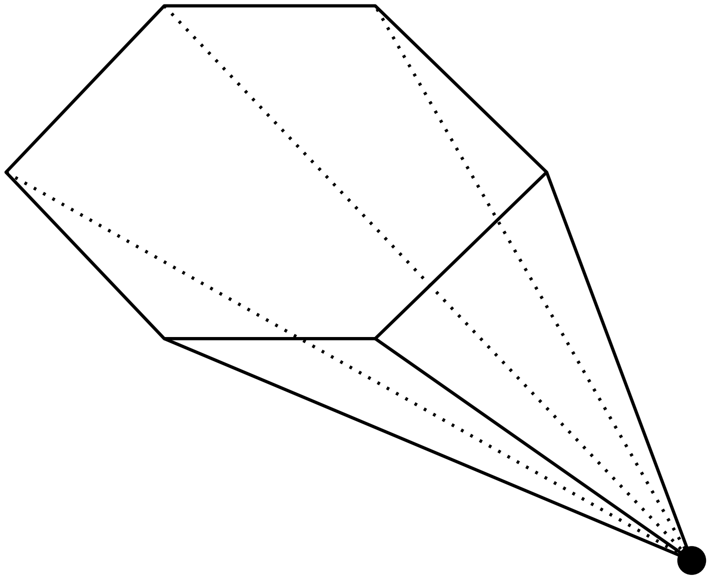

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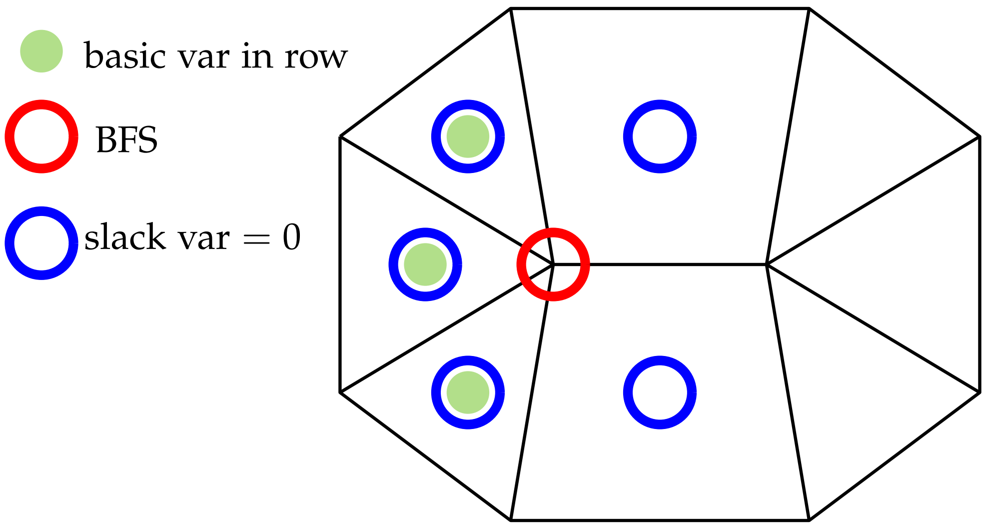
This can make the simplex method go to the next basis **without** increasing the objective value (i.e. because it is stuck inside the same basic feasible solution)

$\rightsquigarrow$ stalling, or even worse cycling

Degeneracy: Stalling

- **Stalling** (taking several iterations to improve the objective function) is sometimes inevitable. Depending on the LP you are trying to solve it can be more or less of a problem.



view from above with the horizontal line "sticking out"

direction of objective function $\longrightarrow$

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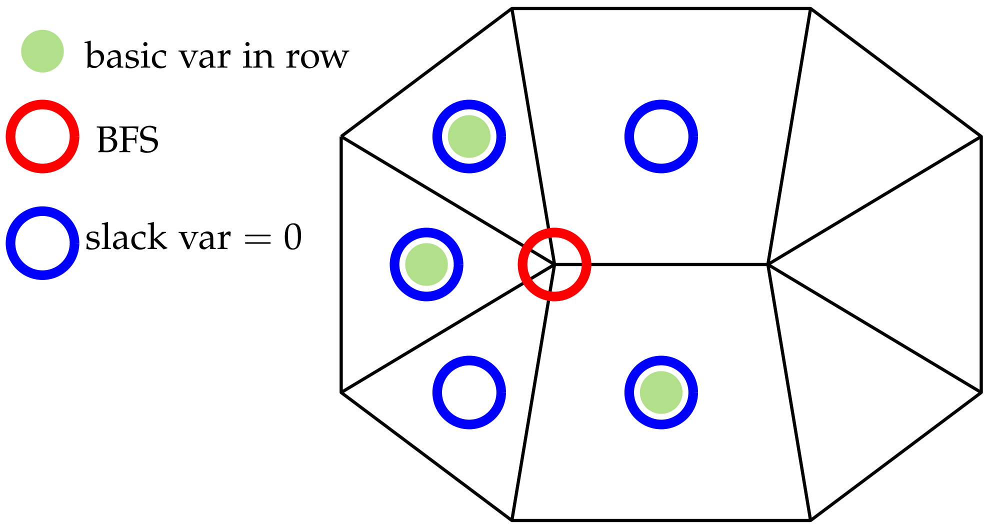

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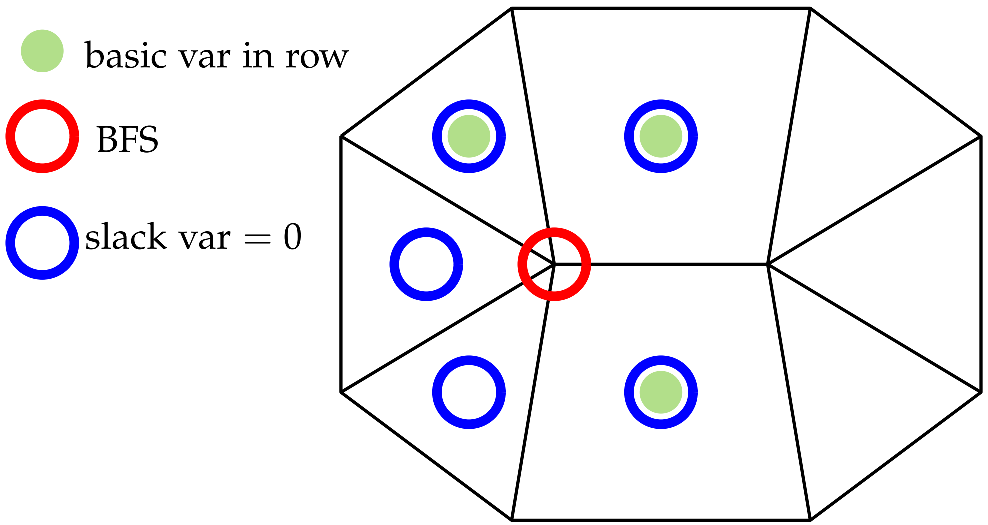

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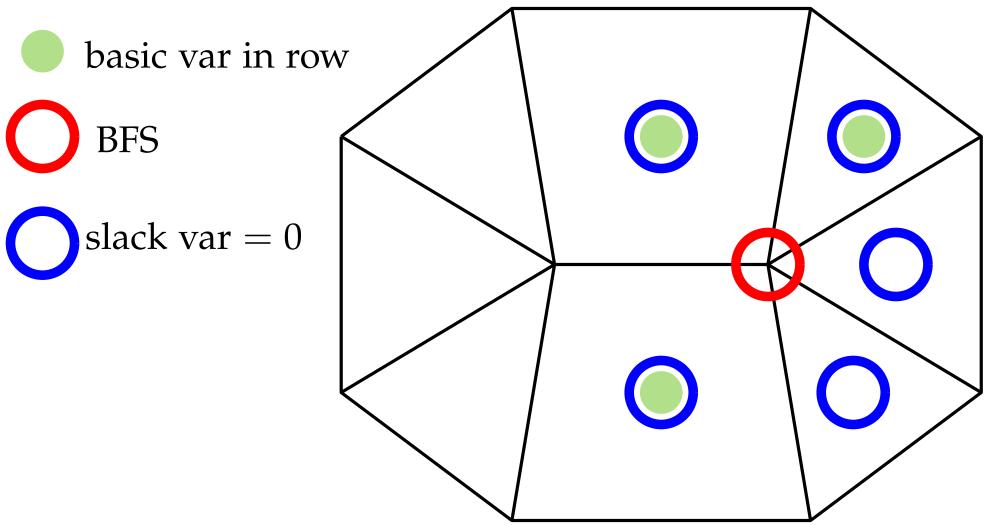

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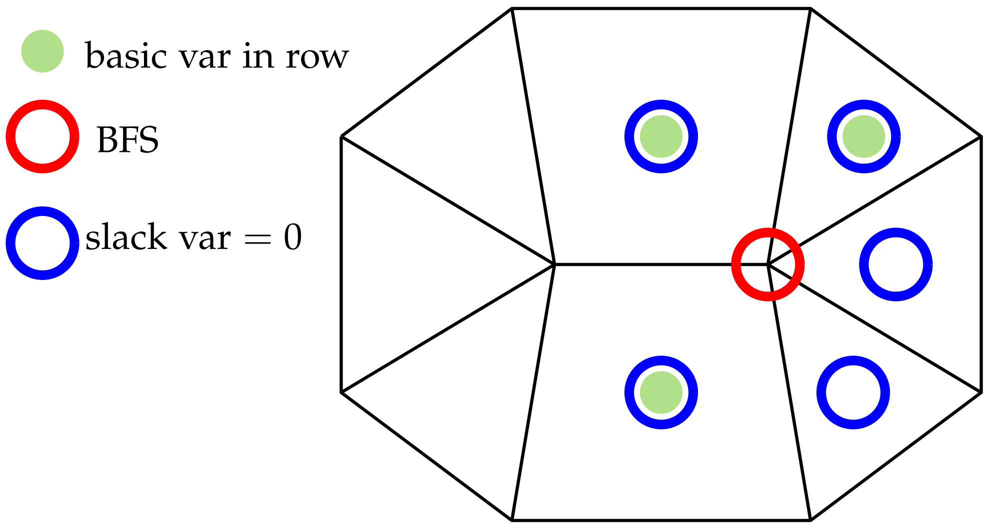

view from above with the horizontal line "sticking out"

direction of objective function $\longrightarrow$

- The objective function only improved after the third pivot!

Degeneracy: Algebraic View

- you can "see degeneracy coming" in the simplex tableau.
- Suppose we have a tie for the leaving variable

basic	Z	x_1	x_2	x_3	x_4	x_5	x_6	*RHS*
			-7			0	0	.
.						0	0	.
x_3			2			1	0	6
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x_4			4			0	1	12
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.						0	0	.
x_3			0			1	$-1/2$	0
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x_2			1			0	$1/4$	3
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x_3 is now in the basis with value 0

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- when x_3 leaves the basis, the "0" means that the objective function does not change!

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- ... it depends, e.g., the structure of your LP might be vulnerable to degeneracy (many constraints tight at once)
- If a Simplex LP solver is stalling regularly and badly, this can be the right moment to switch to an interior-point solver!
- We can't prove that the Simplex algorithm is an efficient (i.e. polynomial-time) algorithm (even in practice it is very fast). *All pivot rules* (so far) have pathologically bad examples

Degeneracy: Cycling??

- Possible in theory to loop around BFSs ...

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(such cycle-avoiding rules tend to be slow in practice ...)

Recap and Properties

1. If there is **exactly one** optimal solution, then it must be a CPF solution.
2. If there are multiple optimal solutions (and a bounded feasible region), then at least two must be **adjacent CPF** solutions.
3. There are only a **finite number** of CPF solutions.
4. If a CPF solution has no adjacent CPF solutions that are better (as measured by Z), then there are **no better CPF** solutions anywhere.